

ESP32-S3: real IQ over USB into SDR++ and SDR#

ESP-SDR firmware, IQS mode (branch s3-iq-stream) · esp-sdr-bridge (SpyServer + rtl_tcp) · Signal Hound VSG60 · 2026-10-02

Turbo Mode developed by Zoltan Doczi from <https://www.z2labs.io>

91 min

continuous SpyServer soak,
18 chaos blocks (hops,
rate/format/gain switches,
reconnects)

0

SpyServer sequence gaps; 0
CRC errors in ~2.5 M IQS1
frames

58.0 dB

tone SNR per bin, -50 dBm
CW at 2350 MHz, 250 kS/s
int16

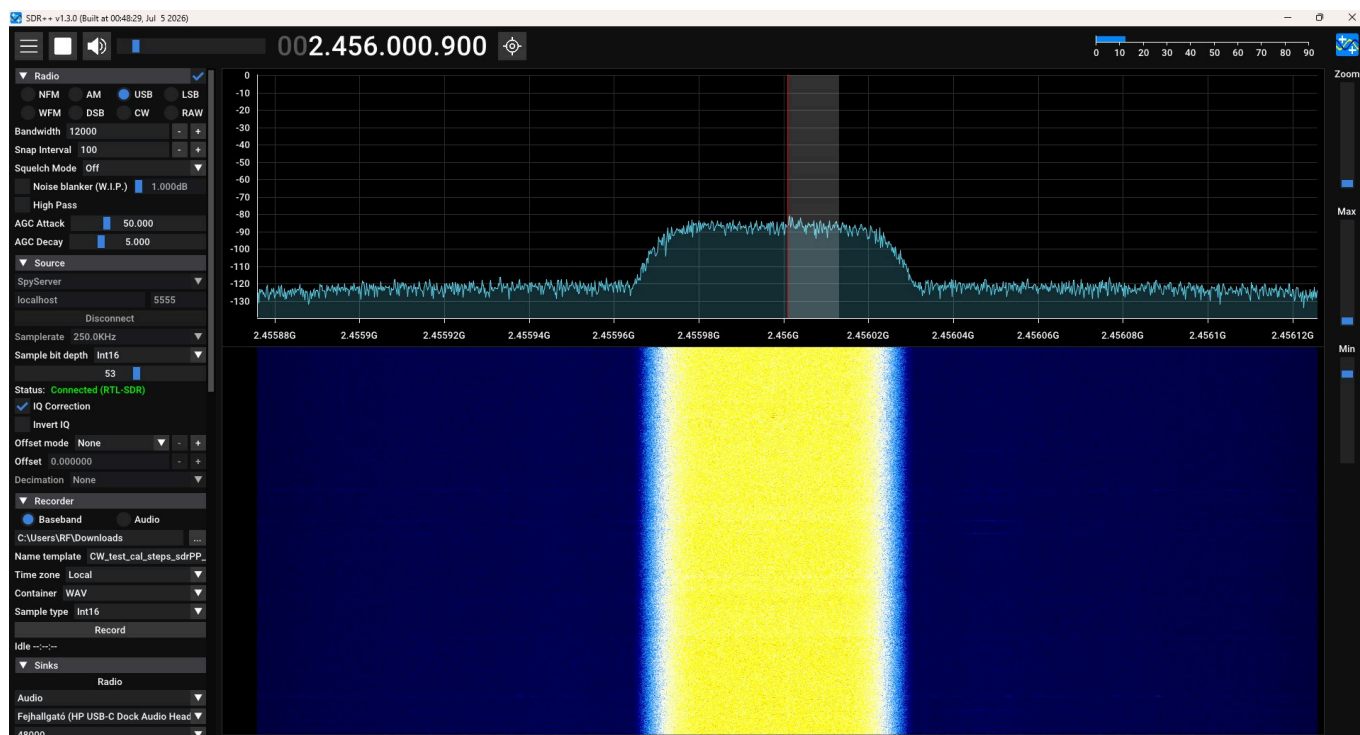
-1.07 ppm

frequency error (crystal),
stable to ± 0.02 ppm after
warm-up

The ESP32-S3 streams gapless, decimated complex baseband (250 / 125 / 62.5 kS/s) over its USB Serial/JTAG port. A small host bridge serves it as SpyServer and rtl_tcp, so SDR++ (also on Android over the LAN), SDR#, gqrx and GNU Radio work without plugins. Tuning range 2204-2804 MHz in 1 kHz steps.

Signal chain

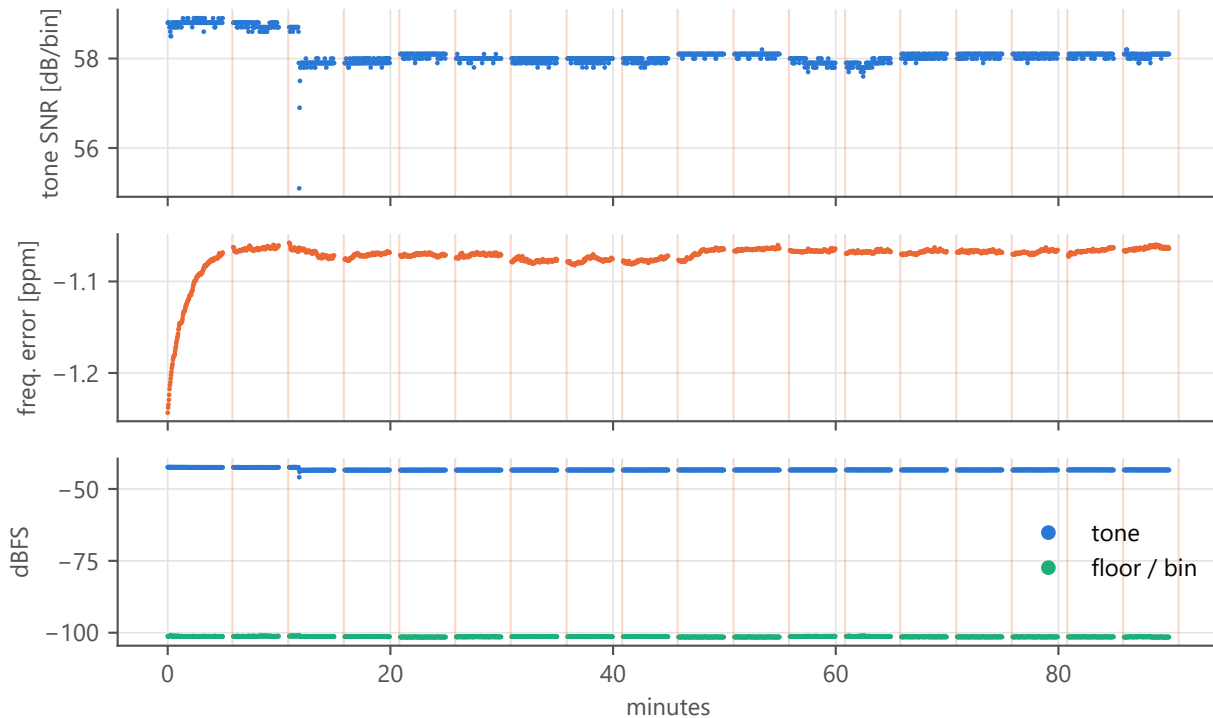
- **Firmware (core 1, PIE SIMD):** 16 MS/s ring → split I/Q int16 (ee.vunzip.16) → optional +fs/4 rotation (LO tuned 4 MHz below the band, so LO leakage and the 1/f hump are filtered out) → FIR /8, 32 taps (triangle × Kaiser, double zeros at k·fs/8) → FIR /(D/8), up to 256 taps, Kaiser β 5.65 → round-to-nearest, 8 or 16 bit → IQS1 frames (64-bit sample index, CRC32). ~12 CPU cycles per input sample pair.
- **Link:** 250 kS/s with 8-bit samples (0.5 MB/s, TPDF dither in the bridge for rtl_tcp), 125 / 62.5 kS/s with 16-bit samples. The USB Serial/JTAG payload limit is ~0.65 MB/s.
- **Bridge:** checks CRC and sample-index continuity, removes residual DC, keeps samples in FIR units (same level for every rate and format), one sender thread and queue per client so a slow client never stalls the USB reader.



SDR++ via the SpyServer bridge, 50 ksym/s QAM16 from the VSG60 at 2456 MHz.

Soak test, 91 minutes

2350 MHz (outside the ISM band), CW 40 kHz above, -50 dBm, int16 at 250 kS/s, gain index 60, back-to-back 2 s windows (8192-point Hann Welch, 30.5 Hz bins). The test client sends exactly the SpyServer commands SDR++ sends. Every 5 minutes a chaos block (orange lines: client reconnect).



Steady state (2229 windows)	p1	median	p99
Tone SNR [dB/bin]	57.8	58.0	58.8
Tone level [dBFS]	-43.5	-43.4	-42.4
Floor [dBFS/bin]	-101.5	-101.4	-101.1
Frequency error [ppm]	-1.17	-1.07	-1.06

- **Integrity:** 0 CRC errors, 0 SpyServer sequence gaps, 0 messages dropped by the bridge. The firmware flagged 8 dropped-frame events (38,912 samples, 0.003 % of the stream) in two clusters 90 s apart: the PC paused reading USB and the S3 TX queue overflowed. Each such gap is visible to the host from the sample index. The same happened during the later CW sweep exactly when the PC was busy with other jobs (git, PDF rendering): a few more such events and 1 CRC error, all detected. A larger host-side read buffer / higher reader thread priority is the next improvement.
- **Frequency:** a constant crystal offset of this board (-1.07 ppm, use `--ppm 1.07`), 0.18 ppm warm-up in the first 5 minutes.
- At ~12 minutes the tone level stepped down by 1.0 dB and stayed there while the floor did not move: a change in the signal path (VSG or RF), not in the stream.

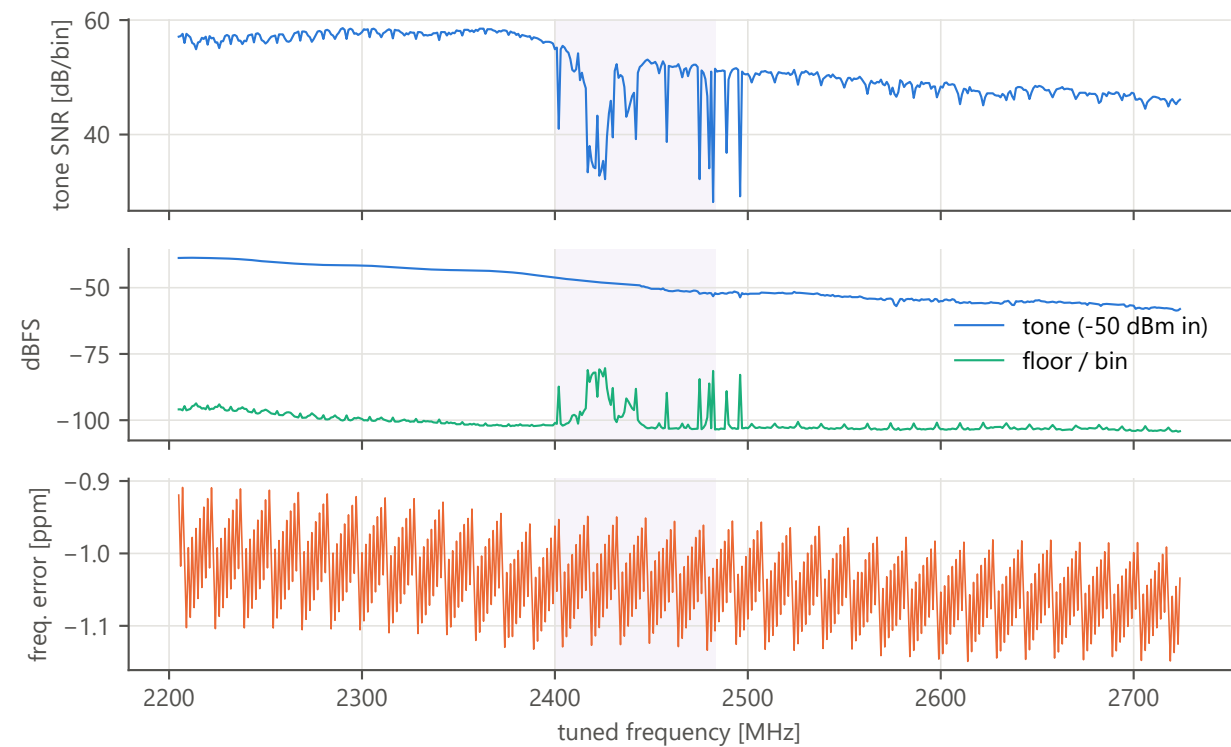
Chaos blocks (median per step)

Step	n	rate [kS/s]	SNR [dB/bin]	tone [dBFS]	floor [dBFS/bin]
125 kS/s int16	18	125.7	61.4	-43.4	-104.8
62.5 kS/s int16 (tone +15 kHz)	18	62.5	64.9	-43.4	-108.3
250 kS/s uint8	18	256.0	56.8	-43.4	-100.2
250 kS/s float	18	256.4	58.0	-43.4	-101.4
gain index 40	18	255.9	44.9	-63.6	-108.5
gain index 70	18	256.2	58.8	-33.9	-92.7
after client reconnect	18	256.4	58.1	-43.4	-101.5
random hops 2210-2790 MHz	54	250.5	50.4	-51.2	-102.3
VSG off (floor only)	18	251.2	-	-	-101.6

Rate: wall-clock estimate per window, reads ~1 % high (timing starts after a USB burst). Level and floor per bin scale with the bin width, so 125k and 62.5k show a 3 / 6 dB lower floor at equal noise density.

CW sweep, 1 MHz steps (520 points, 2205-2724 MHz)

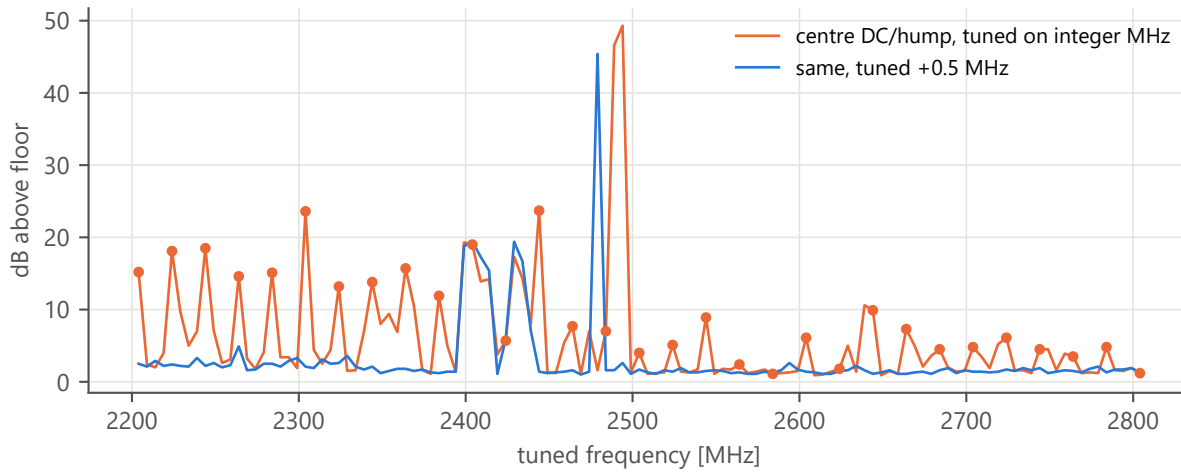
SpyServer int16, 250 kS/s, gain index 60, CW -50 dBm 40 kHz above each tuned frequency, 1 s per point after 1.5 s settling. Purple band: 2.4 GHz ISM.



Band	n	SNR median	SNR min	tone [dBFS]	floor [dBFS/bin]	error [ppm]
2205-2399 MHz	195	57.5	54.9	-41.7	-99.7	-1.02
2400-2484 MHz	85	51.0	28.2	-49.0	-101.0	-1.04
2485-2599 MHz	115	49.8	29.2	-53.0	-103.0	-1.05
2600-2699 MHz	100	47.3	45.1	-55.7	-103.2	-1.06
2700-2724 MHz	25	46.1	44.5	-57.7	-104.0	-1.07

- At a fixed gain index the receive gain falls smoothly by ~17 dB from 2.2 to 2.65 GHz; the SNR follows it (front-end response).
- SNR dips and floor spikes in 2.40-2.50 GHz: live Wi-Fi/Bluetooth traffic in the lab.
- Frequency error: crystal offset plus a ± 0.1 ppm sawtooth repeating every few MHz (PLL fractional resolution with 1 kHz offset steps), below ± 250 Hz at 2.4 GHz.

PLL spur hump at the band centre (rtl_tcp path, 5 MHz steps, VSG off)



When the LO (tuned frequency - 4 MHz) falls on a 20 MHz grid (dots), a hump of close-in spurs (2-4 kHz) appears at the centre, strongest in 2.2-2.4 GHz; tuning 0.5 MHz away removes it. A host-side fix is to avoid those LO values and shift the DDC.

Reproduce

- Firmware: github.com/zodoczi/esp-sdr, branch `s3-iq-stream` (one commit on top of ESPARGOS/esp-sdr main).
- Bridge and test client: github.com/z2labs/esp-sdr-bridge (`esp-sdr-bridge --port COM4; tools/ss_client.py`).

ESP-SDR by the ESPARGOS project. Measurements by Zoltan Doczi, z2labs.